CUNEF UNIVERSIDAD

Quantitative Methods for Humanities

Probability

2024/2025

Basic notions of Probability

What is Probability?

- **Measure of Uncertainty:** Quantifies how likely an event is to occur.
- **Examples.** How likely is:
  - for Obama to win the 2008 US presidential election?
  - for people living in a certain region to belong to a given racial group?
  - to pass a quiz without studying, by answering at random? :)
  - to suffer from a drug side effect?

Frequentist Interpretation

- **Definition:** Probability = Relative frequency in repeated experiments under identical conditions.
- **Example:** A fair coin lands heads $\approx 50\%$ over many tosses.

Challenges:

- Experiments can't always be repeated identically or infinitely.
- Social/political events (e.g., elections) don't fit this framework.

Bayesian Interpretation

- **Subjective Belief:** A measure of one's confidence about an event.
- Values range from 0 (impossible) to 1 (certain).
- **Example:** The probability of Obama winning in 2008 = X%, reflects belief.

Frequentists vs Bayesians

- **Frequentist:** Objective, based on repeatable experiments.
- **Bayesian:** Subjective, based on individual belief and prior knowledge.

Critiques and Counterpoints

- **Critics:** Subjectivity hinders scientific consensus.
- **Bayesians:** Recognizing subjectivity makes science more honest.

Definitions and axioms

Unified Mathematical Foundation

Despite the apparent discrepancies between Frequenstist and Bayesians,

- both frameworks share the same mathematical rules;
- they are unified in the common Theory of Probability, developed by Andrey Kolmogorov in 1933;
- differences are only interpretative.

Basic concepts

- **Experiment:** An action or set of actions producing stochastic events of interest.
- **Sample Space (S):** The set of all possible outcomes of an experiment.
- **Event:** A subset of the sample space.

Examples:

- Flipping a coin: $S = \{\text{Heads, Tails}\}$
- US Presidential Election: $S = \{\text{Obama wins, McCain wins, Third-party wins}\}$

Definition of Probability

For equally likely outcomes, the probability of an event A is:

$$P(A) = \frac{\text{Number of elements in } A}{\text{Number of elements in } S}$$

Example:

- Tossing a fair coin 3 times: $S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$
- Event A: Landing on heads at least twice $A = \{HHH, HHT, HTH, THH\}$
- $P(A) = \frac{4}{8} = 0.5$

Probability Axioms

1. **Non-negativity:** $P(A) \geq 0$
2. **Normalization:** $P(S) = 1$
3. **Addition Rule (Mutually Exclusive Events):** $P(A \text{ or } B) = P(A) + P(B)$ if $A \cap B = \emptyset$.

Example of mutually exclusive events:

- Event A: Obama wins
- Event B: McCain wins
- There is no third party
- $P(A \text{ or } B) = P(A) + P(B)$

General Addition Rule

For events A and B (not mutually exclusive):

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

Example:

- Event A: Obama wins
- Event B: McCain wins
- There is a third party
- $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

Complement Rule and Total Probability

- **Complement Rule:** $P(A^c) = 1 - P(A)$
- **Law of Total Probability:** $P(A) = P(A \text{ and } B) + P(A \text{ and } B^c)$

Example:

- $P(A^c)$: Probability of not A
- $P(A)$: Decomposed into contributions from B and B^c

Counting to compute probability

Permutations

- **Permutations:** The number of ways to arrange k objects out of n unique objects.
- **Example:**
  - How to arrange 2 objects out of 3 unique objects?
  - Objects: A, B, C
  - Arrangements: $\{AB, AC, BA, BC, CA, CB\}$
  - Total number of arrangements: $3! = 3 \times 2 \times 1 = 6$
- **General formula:** ${}_nP_k = \frac{n!}{(n-k)!}$
  - $n!$ is the factorial of n, defined as $n! = n \times (n-1) \times \cdots \times 1$

The Birthday Problem

- **Question:** How large should a group be so that the probability of at least two people in the group sharing the same birthday exceeds 0.5?
- **Key Idea:**

$$P(\text{at least two people have the same birthday}) = 1 - P(\text{nobody shares a birthday})$$

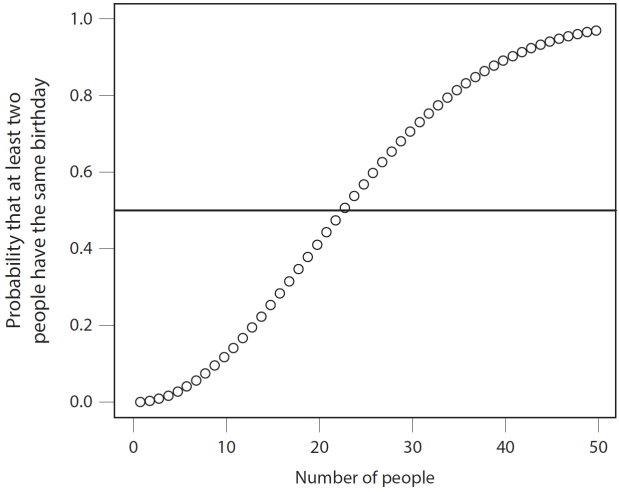
- **Formula for P(nobody shares a birthday)**

$$P(\text{nobody shares a birthday}) = \frac{{}_{365}P_k}{365^k} = \frac{365!}{365^k(365-k)!}$$

- **How to compute?** $365!$ and 365^k are large numbers. Try to compute them in your calculator
- **The log solution:** Work in logarithmic scale (products transform into sums) and then exponentiate back to the original scale
- **in R:** `lfactorial`

  Up to +0.2 points for the first student posting in the forum a piece of code with a function to compute the probability asked, provided the number of people

Solution to the Birthday Problem



Combinations

- **Combinations:** Ways to choose k distinct objects from n elements **without considering order**.
- **Difference from permutations:** Order does not matter.
- **Example:**
  - **Objects:** A, B, C
  - **Permutations (order matters):** $AB, BA, AC, CA, BC, CB \Rightarrow 6$
  - **Combinations (order doesn't matter):** $AB, AC, BC \Rightarrow 3$
- **General formula:** ${}_nC_k = \binom{n}{k} = \frac{n!}{k!(n-k)!}$

Example: probability of choosing at least 2 women in the committee

Problem: Suppose that we are creating a committee of 5 out of 20 people (10 men and 10 women). Each person is equally likely to be assigned to the committee. What is the probability that at least 2 women are on the committee?

Solution:

- Use the complement event and the additivity of the probability for disjoint sets:

$$P(\text{at least 2 women}) = 1 - P(\text{no woman}) - P(\text{exactly 1 woman})$$

- **Total ways to form a committee of 5 from 20:** $\binom{20}{5} = 15,504$
- **Total ways to choose no woman (5 men):** $\binom{10}{5} = 252$
- **Total Ways to choose exactly 1 woman:** $\binom{10}{1} = 2100$
- **Compute probability:**

$$P(\text{at least 2 women}) = 1 - \frac{252}{15,504} - \frac{2,100}{15,504} = 0.84.$$

Example. Acrostic by change?

California Governor Arnold Schwarzenegger's Veto Message in 2009.

To the Members of the California State Assembly:

I am returning Assembly Bill 1176 without my signature.

For some time now I have lamented the fact that major issues are overlooked while many unnecessary bills come to me for consideration. Water reform, prison reform, and health care are major issues my Administration has brought to the table, but the Legislature just kicks the can down the alley.

Yet another legislative year has come and gone with out the major reforms Californians overwhelmingly deserve. In light of this, and after careful consideration, I believe it is unnecessary to sign this measure at this time.

Sincerely,

Arnold Schwarzenegger

- **Question:** Was the "Fuck you" acrostic intentional?
- **Schwarzenegger's reply:** "My goodness. What a coincidence. I suppose when you do so many vetoes, something like this is bound to happen."

Solution to the Veto Message Problem

- The possible line breaks for each given letter are as follows:
  - "u" has **1** possible break (in "unnecessary").
  - "c" has **3** possible breaks (in "come," "consideration," "care").
  - "k" has **1** possible break (in "kicks").
  - "y" has **2** possible breaks (in "Yet," "year").
  - "o" has **2** possible breaks (in "overwhelmingly," "of").
  - "u" again has **1** possible break (in "unnecessary").
- Multiplying these possibilities:

$$1 \times 3 \times 1 \times 2 \times 2 \times 1 = 12$$

- Thus, there are **12 different ways** to create these line breaks.

Solution to the Veto Message Problem

- Total number of ways to insert 6 line breaks into 85 words (`choose(84, 6)` in R):

$$\binom{84}{6} = \frac{84!}{6! \cdot 78!} \approx 406,481,544$$

- **Probability:**

$$P = \frac{12}{\binom{84}{6}} \approx \frac{12}{406,481,544} \approx 1 \text{ in } 34 \text{ million}$$

- **Conclusion:** The acrostic was an extremely unlikely event.

Conditional, Marginal, Joint Probabilities, and Independence

Conditional probability as a way to update our believes

- Conditional probability allows us to **update our beliefs about the likelihood of an event based on new information.**
- It refines our predictions by incorporating relevant evidence, making probability assessments more accurate.
- **Bayes' Theorem** is a fundamental tool that uses conditional probability to update beliefs and make informed decisions.

> How can we live if we don't change?
>
> - Beyoncé. Lyric from "Satellites."

Our brains use conditional probability all the time. Example 1

Example: Checking the Weather Before Going Outside.
Suppose you're about to leave your house, and you quickly glance out the window to check the weather.

- **Initial Belief:** You checked the forecast earlier, and it said there was a 30% chance of rain.
- **New Information:** You now see dark clouds forming and feel a sudden drop in temperature.
- **Refined Prediction:** Your brain updates the probability of rain, now considering it much higher. You grab an umbrella before heading out.

Even without formal calculations, your brain is constantly updating expectations based on incoming information, just like conditional probability!

Our brains use conditional probability all the time. Example 2

Example: You are meeting a friend for coffee, and before you arrive, you wonder what mood she will be in.

- **Initial Belief:** You have a general idea of your friend's mood based on past experiences.
- **New Information:** When you meet her she is smiling and speaking energetically.
- **Refined Prediction:** You update your belief that she is likely to be in a good mood.

Even without formal calculations, your brain is constantly updating expectations based on incoming information, just like conditional probability!

Definitions

- **Conditional Probability:** The probability of event A given B has occurred.

$$P(A|B) = P(A \text{ and } B)/P(B)$$

- **Joint Probability:** The probability of two (or more) events occurring together.

$$P(A \text{ and } B) = P(A \cap B)$$

- **Marginal Probability**[1]**:** The probability of a single event occurring.

$$P(B)$$

[1] The concept of marginal probability implies that some joint probability has been considered previously

Conditional probability. Problem

Scenario: Consider two couples expecting twins. One couple had an ultrasound and determined that at least one twin is a boy. The other couple found out at delivery that the first-born is a boy. What is the probability that both twins are boys in each case?

Solution:

- The sample space for the twins' genders is:

$$\Omega = \{GG, GB, BG, BB\}$$

- **Couple 1: knowing at least one twin is a boy**

$$P(BB|\text{at least one boy}) = \frac{P(BB)}{P(BB \cup BG \cup GB)} = \frac{1/4}{3/4} = \frac{1}{3}.$$

- **Couple 2: knowing the elder twin is a boy**

$$P(BB|\text{elder twin is a boy}) = \frac{P(BB)}{P(BB \cup BG)} = \frac{1/4}{1/2} = \frac{1}{2}.$$

Conclusion: The information upon we condition matters.

Law of Total Probability

Law of Total Probability

The (marginal) probability of an event A can be computed using all possible scenarios B_i:

$$P(A) = \sum_i P(A \cap B_i) = \sum_i P(A|B_i)P(B_i)$$

Example: Probability of fake news

In an online journal, the probability of an article being classified as "fake news" (FN) is 0.29 if its title contains exclamation marks (E), and 0.11 if it does not. If 16% of the articles contain exclamation marks, what is the probability of an article being classified as "fake news"?

Solution:

$$\begin{aligned} P(FN) &= P(FN \mid E)P(E) + P(FN \mid E^c)P(E^c) \\ &= 0.29 \times 0.16 + 0.11 \times 0.84 = 0.1388 \end{aligned}$$

Independence

- Events A and B are **independent** if knowing B gives no additional information about A:

$$P(A|B) = P(A) \quad \text{and} \quad P(B|A) = P(B)$$

- Equivalent Definition:

$$P(A \text{ and } B) = P(A)P(B)$$

The Monty Hall Problem

Problem Setup:

- Choose one of three doors:
  - One door conceals a car (win).
  - Two doors conceal goats (lose).
- Monty opens a door with a goat and offers you the chance to switch.
- **Question:** Should you switch or stay to maximize your chance of winning?

Solution of the Monty Hall problem. Intuition

- **Common (mistaken) belief:** "Now I have fifty-fifty of winning, so why changing?"
- **You don't have "fifty-fifty":**
  - **If you don't change your original choice:**
    When your choice was made you had one-in-three probability of winning. If you stick to your original election, revealing a door does not change that probability.

$$P(\text{car} \mid \text{do not switch}) = 1/3$$

  - **If you change your original choice:**
    In this case you would win the car whenever you would have been mistaken, which has two-in-three probability of occur.

$$P(\text{car} \mid \text{switch}) = 2/3$$

See a more intuitive explanation here.

Solution of the Monty Hall problem. Law of total probability

- Probability of winning if you:
  - **Do not switch:**

$$
\begin{aligned}
P(\text{car}) &= P(\text{car} \mid \text{car first})P(\text{car first}) + P(\text{car} \mid \text{goat first})P(\text{goat first}) \\
&= P(\text{car first}) = \frac{1}{3}.
\end{aligned}
$$

  - **Switch:**

$$
\begin{aligned}
P(\text{car}) &= P(\text{car} \mid \text{car first})P(\text{car first}) + P(\text{car} \mid \text{goat first})P(\text{goat first}) \\
&= P(\text{goat first}) = \frac{2}{3}.
\end{aligned}
$$

- **Not independence between winning and changing your mind:** The event "wining the car" is not independent to the event "switching the door".

Up to +0.2 points for the first student posting in the forum a piece of code with a function that computes the probabilities of winning and losing of the Monty Hall problem based on simulations

Bayes's rule

Bayes' Rule: Definition

Bayes' Rule

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

- $P(A)$: Prior probability (initial belief).
- $P(A|B)$: Posterior probability (updated belief after observing B).
- $P(B|A)$: Likelihood of observing B given A.
- $P(B)$: Marginal probability of B.

Example: Economic Recession Forecast

Consider the following economic forecasting problem:

- Historically, there is a 1 in 20 chance that the economy will enter a Recession (R) in any given year.
- An economic indicator categorizes the economy of the current year as **High-Risk (HR)** for recession.
- Out of 100 years with a recession, 80 will be identified as high-risk by the indicator, and 20 will not be detected.
- There is a 1 in 10 chance that the economy is not in recession but the indicator still signals high-risk.

Given the result of the economic indicator, what is the probability that the economy will enter a recession?

Economic Recession Forecast

Problem:

- Probability of a recession $P(\mathrm{R}) = 1/20 = 0.05$.
- Economic indicator is high-risk (HR) with:
  - True positive rate $P(HR|R) = 0.80$,
  - False positive rate $P(HR|R^c) = 0.10$.

Bayes' Rule:

$$P(R|HR) = \frac{P(HR|R) \cdot P(R)}{P(HR|R) \cdot P(R) + P(HR|R^c) \cdot P(R^c)}$$

Result:

$$P(R|HR) = \frac{0.80 \cdot 0.05}{0.80 \cdot 0.05 + 0.10 \cdot 0.95} \approx 0.297 \quad (29.7\%)$$

Random variables

What is a Random Variable?

- A random variable assigns a numeric value to each event in the sample space. That is,

$$X : S \to \mathbb{R}.$$

- Values must represent:
  - **Mutually exclusive events:** No overlap between values.
  - **Exhaustive events:** All events are represented.
- Examples:
  - Coin flip: Heads = 1, Tails = 0.
  - Racial group: Black = 1, White = 2, Hispanic = 3, etc.

Types of Random Variables

- **Discrete Random Variable:**
  - Takes a finite or countably infinite set of values.
  - Examples: Number of years of education, racial group.
- **Continuous Random Variable:**
  - Takes any value within an interval of the real line.
  - Examples: Height, weight, GDP.

Mass, density, and distribution

- **Probability Mass Function (PMF):**
  - Defines the probability for discrete random variables to take a single value.
  - **Definition:** $f(x) = P(X = x)$.
- **Probability Density Function (PDF):**
  - Defines the likelihood for continuous random variables.
  - It is a **positive function whose total area under the curve equals 1**, that is

$$f(x) \geq 0, \quad \int_{\mathbb{R}} f(x)\,\mathrm{d}x = 1.$$

- **Cumulative Distribution Function (CDF):**
  - Represents the probability that a random variable is less than or equal to a value x.
  - **Formula:**

$$F(x) = P(X \leq x) = \begin{cases} \sum_{k \leq x} f(k) & \text{(discrete)} \\ \int_{-\infty}^{x} f(t)\,dt & \text{(continuous)} \end{cases}$$

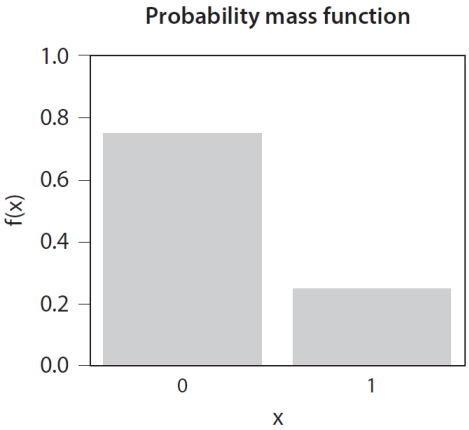
Bernoulli Distribution. The simplest discrete distribution

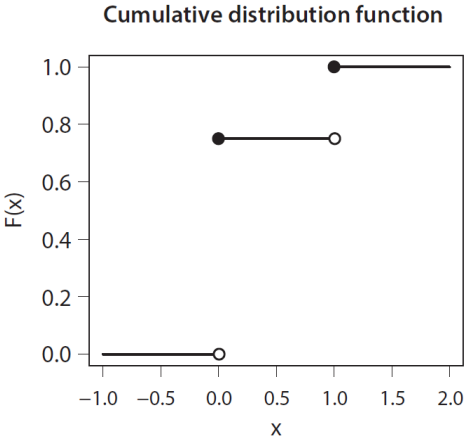
- **Definition:** A discrete random variable representing two outcomes (e.g., success and failure).
- **PMF:**

$$f(x) = \begin{cases} p, & \text{if } x = 1 \\ 1 - p, & \text{if } x = 0 \end{cases}$$

- **Examples:**
  - Coin flip: $P(\text{Heads}) = p$, $P(\text{Tails}) = 1 - p$.
  - Voter turnout: $P(\text{Vote}) = p$, $P(\text{Abstain}) = 1 - p$.

Visualization of the Bernoulli distribution





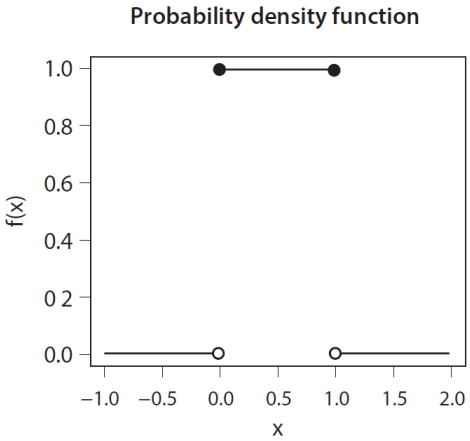
Uniform Distribution. The simplest continuous distribution

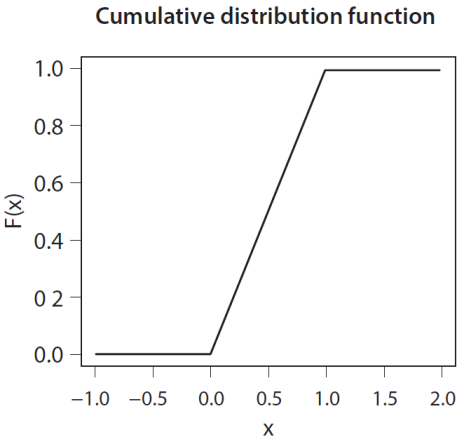
- **Definition:** A continuous random variable with equal probability for all values in an interval $[a, b]$.
- **PDF** (`dunif()`):

$$f(x) = \begin{cases} \frac{1}{b-a}, & \text{if } a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$$

- **Example:**
  - Randomly choosing a point in $[0, 1]$.

Visualization of the uniform distribution

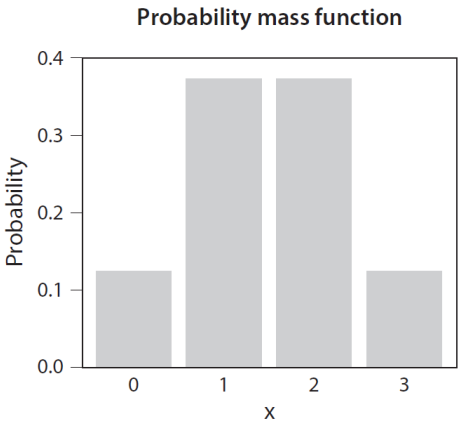


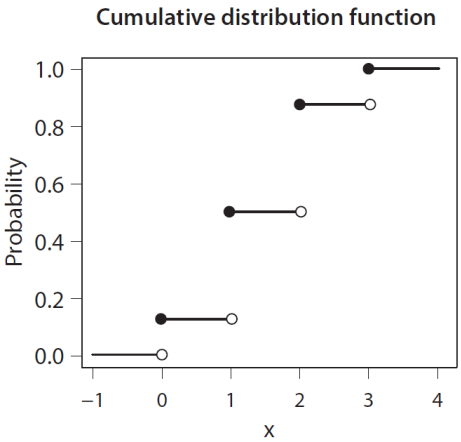


Binomial Distribution

- **Definition:** A binomial random variable X represents the number of successes in n **independent trials**, each with success probability p.
- **Examples:**
  - Number of heads in n coin flips.
  - Number of votes for a candidate in an election.
- **Properties:**
  - X takes values from 0 to n.
  - X is the sum of n **independent Bernoulli** random variables.

Visualization of the binomial distribution





Mass and distribution

- **PMF** (`dbinom()`):

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \ldots, n$$

  - **Recall:** $\binom{n}{x} = \frac{n!}{x!(n-x)!}$ is the number of ways to choose x successes in n trials.
  - **Example:** Probability of 2 successes in 3 independent trials with $p = 0.5$:

$$P(X = 2) = \binom{3}{2} 0.5^2 (1-0.5)^1 = 3 \cdot 0.25 \cdot 0.5 = 0.375$$

- **CDF** (`pbinom()`):

$$F(x) = P(X \leq x) = \sum_{k=0}^{x} \binom{n}{k} p^k (1-p)^{n-k}$$

  - **Example:** Probability of at most 1 success in 3 trials with $p = 0.5$:

$$F(1) = P(X = 0) + P(X = 1) = 0.125 + 0.375 = 0.5$$

Application of Binomial Distribution

- **Election Example:** Probability of a tie when N voters turn out, each with 50% support for a candidate:

$$P(\text{Tie}) = P(X = 500), \quad X \sim \text{Binomial}(n = 1000, p = 0.5)$$

- **R code:**

```
## number of voters who turn out
voters <- c(1000, 10000, 100000)
dbinom(voters / 2, size = voters, prob = 0.5)
## [1] 0.025225018 0.007978646 0.002523126
```

The normal distribution

The Normal Distribution

- Also known as the **Gaussian Distribution**.
- A continuous distribution over the real line $\mathbb{R}$.
- Characterized by two parameters:
  - Mean (μ): Center of the distribution.
  - Standard deviation (σ): Controls the spread.
- Notation: $X \sim N(\mu, \sigma^2)$.

PDF and CDF

- **PDF** (`dnorm()`):

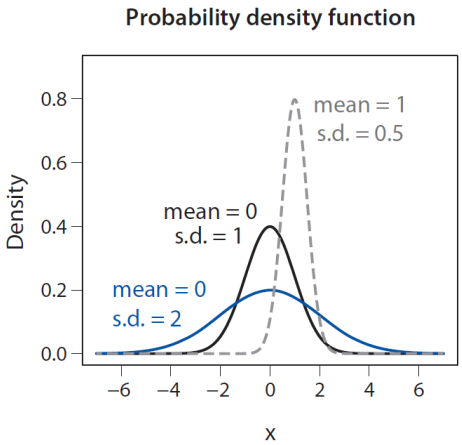
$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

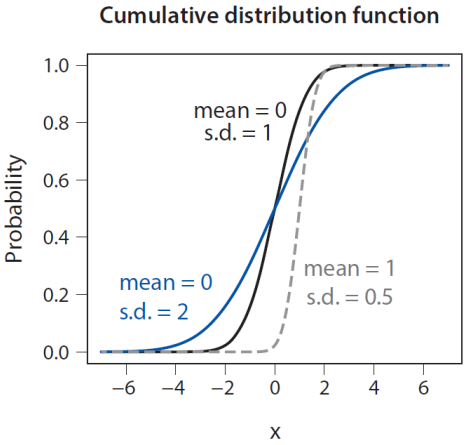
- **CDF:** (`pnorm()`):

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\,dt, \quad \text{no explicit expression}$$

- Symmetric and bell-shaped, centered around μ.
- 68% of data lies within 1σ of μ, and 95% within 2σ.

Visualization of the normal distribution





Properties of the Normal Distribution. Standardization

- Suppose $X \sim N(\mu, \sigma^2)$ is a normal random variable.
- For any constant c, the following properties hold:
  1. Adding c to X:
  $$Z = X + c \implies Z \sim N(\mu + c, \sigma^2)$$
  2. Multiplying X by c:
  $$Z = cX \implies Z \sim N(c\mu, (c\sigma)^2)$$
- **Z-score Transformation:**
$$Z = \frac{X - \mu}{\sigma} \implies Z \sim N(0, 1)$$
- The **Z-score** standardizes X to have zero mean and unit variance.
- **Applications:**
  - Standardizing data for comparison.
  - Calculating probabilities using standard normal tables.
    Useful back in the days but obsolete with the advent of programming languages.

An application

- Based on the 2008 and 2012 US election data, the following model is proposed

$$\begin{aligned}&\text{Obama's 2012 standardized vote share}\\&\quad = 0.983 \times \text{Obama's 2008 standardized vote share} + \epsilon,\end{aligned}$$

  where $\epsilon \sim N(0, 0.18^2)$.
- In 2008, Obama won a standardized vote share of 0.87 in California. What is the probability that Obama wins a greater share of California votes in 2012?
- In 2008, Obama won a standardized vote share of -0.67 in Texas. What is the probability that Obama wins a greater share of Texas votes in 2012?
- **R code:**

```
# probability for California
pnorm(0.87, mean = 0.983 * 0.87, sd = 0.18, lower.tail = FALSE)
# probability for Texas
pnorm(-0.67, mean = 0.983 * -0.67, sd = 0.18, lower.tail = FALSE)
```

Expectation and Variance

Expectation (Mean)

- **Definition:** The theoretical average value of a random variable.
- For a random variable X:

$$E(X) = \begin{cases} \sum_x xf(x), & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} xf(x)\,dx, & \text{if } X \text{ is continuous} \end{cases}$$

- **Examples:**
  - Bernoulli(p): $E(X) = p$
  - Binomial(n, p): $E(X) = np$
  - Uniform[a, b]: $E(X) = \frac{a+b}{2}$

Variance and Standard Deviation

- **Variance:** Measures the spread (from the mean) of a random variable.

$$V(X) = E[(X - E(X))^2] = E(X^2) - (E(X))^2$$

- **Standard Deviation:** The square root of variance:

$$\mathrm{SD}(X) = \sqrt{V(X)}$$

- **Examples:**
  - Bernoulli(p): $V(X) = p(1-p)$
  - Binomial(n, p): $V(X) = np(1-p)$
  - Uniform[a, b]: $V(X) = \frac{(b-a)^2}{12}$

Properties of Expectation and Variance

Expectation Rules:

- $E(a) = a$
- $E(aX) = aE(X)$
- $E(aX + bY) = aE(X) + bE(Y)$
- For independent X and Y: $E(XY) = E(X)E(Y)$

Variance Rules:

- $V(aX) = a^2 V(X)$
- $V(X + b) = V(X)$
- For independent X and Y: $V(X + Y) = V(X) + V(Y)$

Large Sample Theorems

Law of Large Numbers, a.k.a. “The Wisdom of the Crowd”

- In 1906, the statistician Francis Galton[1] observed a competition of eight hundred people trying to guess the weight of an ox at a country fair.
- He discovered that the average guess (1,197lb) was extremely close to the actual weight (1,198lb).
- This illustrates the Law of Large Numbers (LLN): as the number of observations increases, the sample mean approaches the true mean.

[1] The same statistician who ran the first recorded experiment of linear regression

Law of Large Numbers (LLN)

- **Definition:** As the sample size (n) increases, the sample average ($\bar{X}$) converges to the population mean ($E(X)$):

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow[n\to\infty]{} E(X).$$

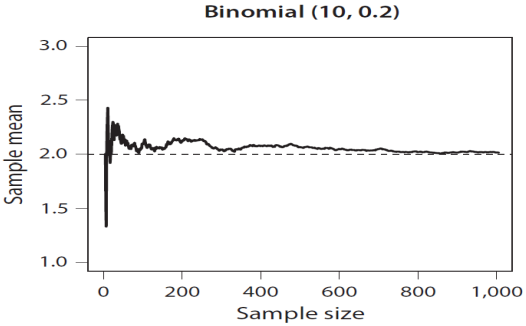
- **Key Idea:** The sample mean becomes a better approximation of the true mean as n grows.
- **Applications:**
  - Survey results approximate population opinions.
  - Simulations approximate theoretical probabilities.
  - Condorcet's Jury Theorem: when certain assumptions hold, the probability that a majority of voters support the correct decision increases and approaches one as the number of voters increases

Illustrating LLN

R code:

```
n <- 1000 # number of simulations
x.binom <- rbinom(n, p = 0.2, size = 10) # simulating
mean.binom <- cumsum(x.binom) / 1:n # computing sample mean
plot(1:1000, mean.binom, type = "l", ylim = c(1, 3), xlab = "Sample size",
     ylab = "Sample mean", main = "Binomial(10, 0.2)")
abline(h = 2, lty = "dashed") # expectation
```

- Simulating $n = 1000$ draws from a Binomial(10, 0.2).
- Sample mean ($\bar{X}$) approaches the expectation $E(X) = 2$.

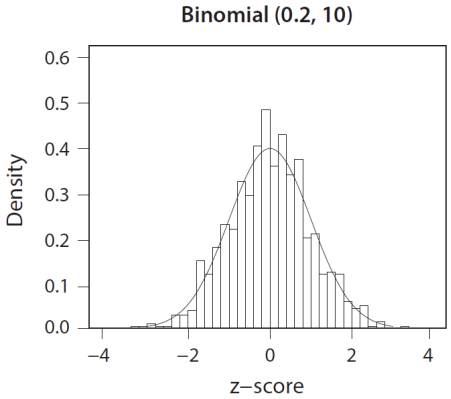
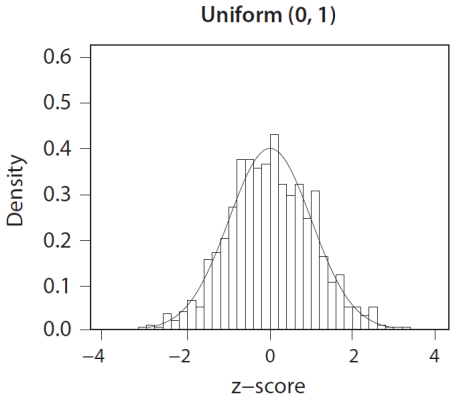


Central Limit Theorem (CLT)

- **Definition:** The sample mean ($\bar{X}_n$) of n independent, identically distributed (i.i.d.) observations converges in distribution to a normal distribution as n increases:

$$Z = \frac{\bar{X}_n - E(X)}{\sqrt{V(X)/n}} \approx N(0, 1)$$

- **Key Idea:** Regardless of the original distribution, the sample mean's distribution becomes approximately normal for large n.
- **Applications:**
  - Estimating error for sample means.
  - Understanding variability in data collection.

Illustrating CLT

- Simulate Z-scores for $n = 1000$.
- Sample mean's z-scores follows $N(0, 1)$ distribution.